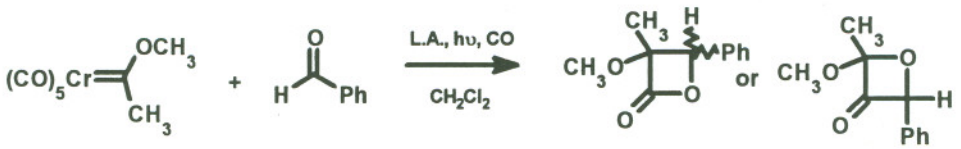
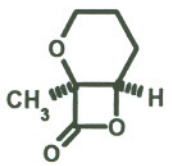
Organic Cume, October 1, 1994

In a number of papers, Hegedus and coworkers have studied the reaction of chromium complexes with unsaturated reagents. (*JACS* **1988**, *10*, 2122; *JACS* **1990**, *112*, 1190; *JACS* **1991**, *113*, 5784; *JOC* **1991**, *57*, 1461; *JACS* **1992**, *114*, 5010; *JOC* **1992**, *57*, 5453; *JOC* **1993**, *115*, 9037; *JOC* **1993**, *58*, 6779; *JOC* **1991**, *56*, 2209; *JACS* **1991**, *113*, 923; *JACS* **1990**, *112*, 4364). In a recent report, (P-J. Colson and L. S. Hegedus *J. Org. Chem.* **1994**, *59*, 4972-76), it was reported that chromium carbene complexes react with aldehydes in the presence of various Lewis Acids (L.A = Lewis Acids) under the conditions shown below to give β-lactones **2**, and 3-oxatanones **4**.



1 **2** **3**

Among other work reported in this paper is the application of this reaction to prepare lactone **4**.



4

1. How does one prepare the starting complex **1** from $Cr(CO)_6$?
2. Give the mechanism for the formation of product **2**.
3. Give the structure for the starting chromium complex that will give **4**.
4. Propose a method for the preparation of the starting chromium complex that gives **4**.